



June 17, 2026

Dear Professor Dinesh Krishnamoorthy,

I am writing to express my strong interest in the PhD position in Data-driven Optimization for Energy Communities at the Norwegian University of Science and Technology (NTNU), located in Trondheim, Norway. The opportunity to contribute to optimizing renewable energy systems using data-driven decision-making aligns perfectly with my research interests and skills.

As a data science engineer with a background in AI, I have developed expertise in building efficient models that drive actionable insights. My most notable achievement is the development of Medical *MambaTS*, a linear-time state space model for sub-millisecond bedside inference. This work has implications for real-time decision-making in energy management systems, where timely optimization can lead to significant efficiency gains.

In particular, my experience with modeling and optimization can be applied to optimizing energy communities by developing data-driven models that predict and manage energy supply and demand. I believe that my knowledge of PyTorch Autograd and 0D Windkessel Physics-Informed Operator Network (PI-DeepONet) can be leveraged to develop a robust framework for energy community optimization.

I am excited about the prospect of joining NTNU's Department of Engineering Cybernetics and contributing to the development of data-driven decision-making tools for energy communities. I believe that my skills and experience make me an ideal candidate for this position, and I look forward to the opportunity to discuss my application with you further.

Sincerely,

Ismail Aïssaoui